WF-Transformer: Learning Temporal Features for Accurate Anonymous Traffic Identification by Using Transformer Networks

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***Abstract*— Website Fingerprinting (WF) is a network traffic mining technique for anonymous traffic identification, which enables a local adversary to identify the target website that an anonymous network user is browsing. WF attacks based on deep convolutional neural networks (CNN) get the state-of-the-art anonymous traffic classification performance. However, due to the locality restriction of CNN architecture for feature extraction on sequence data, these methods ignore the temporal feature extraction in the anonymous traffic analysis. In this paper, we present Website Fingerprinting Transformer (WF-Transformer), a novel anonymous network traffic analysis method that leverages Transformer networks for temporal feature extraction of traffic traces and improves the classification performance of Tor encrypted traffic. The architecture of WF-Transformer is specially designed for traffic trace processing and can classify anonymous traffic effectively. Furthermore, we evaluate the performance of WF-Transformer in both closed-world and open-world scenarios. In the closed-world scenario, WF-Transformer attains 99.1% accuracy on Tor traffic without defenses, better than state-or-the-art attacks, and archives 92.1% accuracy on the traces defended by WTF-PAD method. In the open-world scenario, WF-Transformer has better precision and recall on both defended and non-defended traces. Furthermore, WF-Transformer with a short input length (2000 cells) outperforms the DF method with a long input length (5000 cells).**

***Index Terms*— Website fingerprinting, traffic analysis, encrypted communication, transformer networks.**

I. INTRODUCTION

WITH the development of Internet, privacy disclosure of network users is an urgent problem for network security. Last decades, several anonymous communication systems have been designed and developed for the users‘ privacy protection [1]. Tor (or The Onion Router) is a popular low latency anonymous network, and has more than three million daily users [2]. Latest studies have shown traffic mining algorithm called Website Fingerprinting (WF) technique can destroy the anonymity of the Tor [3]. Since the different add-in and content of different websites, there is a specific pattern of traffic encrypted through an anonymous network system, which can be used for determining the destination of the target website based on the collected traces [4], [5], [6], [7], [8], [9]. Network attackers eavesdrop and collect the network traffic passively when a user visits the target website. WF attackers use the features that are calculated from the network traffic traces, such as packet rates [10], directions of packets [7] and inter-arrive time [11], and then a classifier is trained for the classification of anonymous traces.

The pioneering WF attacks are mainly focusing on the handcrafting feature extraction of the anonymous traces and the selection of classifiers [10], [11], [12], [13], [14]. Wang et al. utilize several traffic features, such as packet direction, packet size, and burst number, and then a k-NN classifier is trained for the prediction of target traces [11]. Panchenko et al. propose a WF attack method named CUMUL [13], which calculates a novel feature set with a cumulative behavioral representation of the collected traffic traces and utilizes Support Vector Machine (SVM) as the classifier. Hayes et al. propose a k-FP method for WF attack [14], which utilizes a random forest to rank the importance of different traffic features and then uses k-NN classifier to predict the labels of target traffic traces. These three WF attack methods achieve about 91% classification accuracy for Tor anonymous traffic trace classification in the closed-world setting.

With the development of WF attack methods, several defense methods have been proposed [15], [16], [17], [18], [19], [20]. Dyer et al. propose a WF defense method named BuFLO [15], which pads a large number of dummy packets into normal traffic traces to protect users’ privacy. Cai et al. improve the BuFLO defense method and propose TAMARAW [16]. TAMARAW sets a hyperparameter to restrict the number of the padded dummy packets during the transmission process, but the bandwidth overhead and the latency are still too high to be implemented in a real network scenario. Juarez et al. propose a lightweight defense method named WTF-PAD

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[17], and the strategy of the WTF-PAD is that it only pads dummy packets into the gaps between the real packets, which decreases the bandwidth overhead efficiently, and has no delay.

However, with the development of deep learning techniques [21], [22], [23], [24], [25], traffic data mining methods that based on CNN network architecture and its variants are used for traffic analysis [26], [27]. WF attacks based on deep convolutional neural networks have been proposed [7], [28], and existing defenses that padding dummy packets into normal trace based on the statistical information of traces fail to counter deep WF attacks [7], [29], [30]. Rimmer et al. test multiple deep frameworks to classify the anonymous traffic and get better performance than existing non-deep WF attack methods [29]. Sirinam et al. utilize a more sophisticated convolutional neural network to construct a deep WF model called DF, and DF only uses the direction of the packets as the input of the model. Both of the deep WF attack methods achieve more than 96% classification accuracy [7].

Although the latest deep WF attack methods based on CNN architecture have obtained a lot of progress in anonymous traffic classification [7], [30], the design of existing deep WF attacks based on CNN fails to extract the temporal feature of the traces, which can be seen as the long-term dependence among the cells of traffic traces. For example, one request is sent to the server by the client, the response of the current request may not be send back to the client immediately due to the response of the former request is transmitting, and the current request has been delayed, so the local features extract by CNN fail to represent the patterns of the traces. CNN only focuses on the local features of the traffic and fails to capture the long-term dependence among the packets in a traffic trace, which can make the discriminative patterns of the traffic traces be ignored, and reduce the performance of the deep model on the classification of anonymous traffic.

To tackle this problem, we present a novel anonymous trace classification model called Website Fingerprinting Transformer (*WF-Transformer*). Unlike the existing deep WF attack models, WF-Transformer utilizes Transformer network [24] to extract the temporal feature of the traffic traces. Furthermore, we propose a more sophisticated design on the network architecture of Transformer and the corresponding optimization function, and make the proposed WF-Transformer more suitable for processing traffic trace data. To demonstrate the effectiveness and superiority of WF-Transformer, we conduct extensive experiments on a public dataset and evaluate the performance of our method in both closed-world and open-world scenarios. Considering the existing WF defense methods, we also conduct experiments to show the effectiveness of our method against WF defenses. To sum up, the contributions of this paper can be summarized as follows:

1) We present *WF-Transformer*, a novel WF attack based on Transformer network. To the best of our knowledge, WF-Transformer is the first WF attack that utilizes Transformer network to extract the temporal feature of the traffic traces for traffic classification, and yields a better performance than state-of-the-art CNN-based WF attacks.

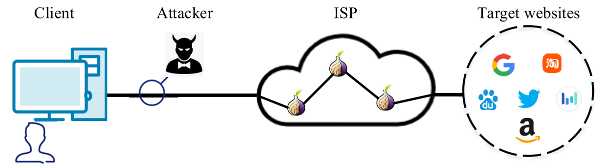


Fig. 1. Demo of website fingerprinting attack.

2) We propose a more sophisticated design of Transformer network for traffic analysis and corresponding optimization function. WF-Transformer shows two main advantages. One is its shorter input than CNN-based methods, which allows an adversary to save time in traffic collection. The other is its fast convergence, which helps to get a robust model.
3) We evaluate the effectiveness of WF-Transformer on the public Tor dataset for both closed-world and open-world settings, and we also show the performance of WF-Transformer on non-defended and defended traces. Compared with the state-of-the-art methods, WF-Transformer outperforms existing methods in terms of classification accuracy in closed-world setting, TPR, FPR, Precision and Recall for open-world setting.
4) We conduct experiments to show the information leakage of features extracted by WF-Transformer and CNN-based model, respectively, and reveal the impact of the Transformer architecture on feature extraction. The study indicates that WF-Transformer can utilize shorter traces to learn more information from traces than CNN-based models with the help of extracted temporal features, demonstrating the superiority and effectiveness of Transformer architecture used for traffic analysis.

The rest of the paper is organized as follows. The threat model about WF attacks is introduced in Section II. Related works are shown in Section III. Preliminary knowledge about Transformer network is introduced in Section IV. The proposed method is shown in Section V. The evaluation setting and experiment results are shown in Section VI and Section VII, respectively. In Section VIII, we show different information leakage for different models. In Section IX, we give some discussion about our proposed method, and we conclude our work in Section X.

II. Threat Model

Tor network is a popular low-latency anonymous system, and encrypted packets are transmitted in Tor with multiple proxies to protect the users' privacy. Figure 1 shows the WF attack scenario, and we can find that attackers are located between the client and the entry node of Tor, and record the traffic traces passively during the user is browsing the target websites with Tor. Generally, the sequence of the packets and time stamps of the trace can be collected, and several handcrafting trace features can be calculated and the classifier can be trained for the identification of the anonymous trace.

We formalize the WF attack scenario in this part. Generally, WF attackers are located between the client and the entry node of the Tor, and they can only record the transmitted packets passively, which means that attackers can not drop, modify, decrypt or delay the packets. Furthermore, the transmitted packets are encrypted with different secret keys in different proxy nodes of Tor. The plaintext information and IP of the packets are invisible to attackers without packet decryption. For the attacker, we assume that the identity of the client is known by the attacker, and the goal of the attacker is to identify the target websites that the user visits.

Closed-World and Open-World Scenario: To evaluate the performance of WF attack, there are two settings, the closed-world setting and the open-world setting. For the closed-world setting, the network users is assumed that they can only access a finite set of websites, and the attacker monitors these websites, and the classification of the anonymous trace is a supervised learning problem. However, this assumption is considered unrealistic, because of the large number of websites in the real-world that can be visited by network users. Furthermore, the traffic traces collected by attackers are limited under this setting. For the open-world setting, which is more practical, Tor users can visit a large number of websites while the attacker can only record a limited of website traces. For a collected trace, attackers need to identify whether the traffic is under monitored. If it is monitored, similar to the closed-world setting, a classification model is trained to identify the target website. The open-world setting is a more complex problem compared with closed-world setting, the number of unmonitored websites is assumed much greater than that of the monitored websites in closed-world setting.

III. Related Works

In this section, we give some surveys on the latest process of WF attack. Considering the influence of WF defenses, the related works are introduced in two parts, WF attacks and WF defenses.

A. WF Attacks

With the wide application of anonymous network, lots of WF attack methods have been proposed [7], [11], [13], [14], [30], [31], [32], [33]. The pioneering attacks are mainly non-deep WF attack methods, which extract the handcrafting features from the collected traffic traces based on human knowledge. Packet direction, packet rates and inter-arrive time of packets are commonly used handcrafting features, and a classifier is trained on these features to identify the target websites of the traffic traces. Existing non-deep WF attack methods aim at trace feature extraction and classifier selection, and KNN and SVM are commonly used classifiers [11], [13], [14], and these WF attack methods achieve 91% classification accuracy in the closed-world setting.

Deep WF attacks aim at utilizing deep neural networks for anonymous trace extraction automatically. Rimmer et al. propose automated website fingerprinting (AWF) which utilizes deep neural networks to identify Tor traffic automatically rather than manually engineering [29]. AWF has tried a lot of state-of-the-art deep networks, such as SDAE, CNN, and LSTM. With the help of deep learning techniques, AWF achieves 96% classification accuracy on Tor traffic in closed-world setting. Sirinam et al.propose Deep Fingerprinting (DF) to classify the anonymous traffic [7]. DF utilizes a CNN network with a sophisticated design. DF only uses the direction of packets as inputs, and an end-to-end training manner is used for the training of the model. In closed-world setting, DF can obtain 98% classification accuracy on non-defended Tor traffic. As for the Tor traffic defended by WTF-PAD method, DF can still achieve over 90% classification accuracy. Bhat et al. propose Var-CNN method [30], which utilizes deep techniques for large amounts of traffic classification. Unlike prior deep WF attacks, VarCNN utilizes time stamp and packet direction as network inputs, and obtains over 98% classification accuracy on Tor traffic [30]. Wang et al. ensemble multiple CNN-based networks for the classification of anonymous traffic [34], and get better performance in open-world scenario.

B. WF Defenses

With the development of WF attacks, several WF defenses have been proposed [15], [17], [35], [36]. The prior defense methods pad dummy packets into normal traces based on statistical information of the Tor traffic to disturb the WF attack classifiers. However, existing defenses, BuFLO [15] and TAMARAW [16], pad too many dummy packets into normal traces and cause high bandwidth overhead, and this makes them impractical.

Several lightweight defenses have been proposed [17], [35], [37]. Juarez et al. propose Website Traffic Fingerprinting Protection with Adaptive Defense (WTF-PAD), and WTF-PAD builds a histogram of packet time from the collected Tor traces. By sampling packet time from the histogram, WTF-PAD only pads dummy packets into the gaps of the normal trace, which reduces the bandwidth overhead. However, WF attacks based on deep learning techniques, such as DF method, still can classify the traffic traces defended by WTF-PAD. Gong et al. propose a lightweight defense called FRONT [35]. FRONT only pads the front portion of Tor traces, which is a feature-rich part of a traffic trace. With similar bandwidth overhead, FRONT gets better performance than WTF-PAD.

Deep classifiers are vulnerable to adversarial examples [38], [39], and several WF defense methods based on adversarial training manners are proposed for against deep WF attacks. Rahman et al. propose Mockingbird method for WF defense, which generates traces by moving the traffic traces randomly and not following more predictable gradients [40]. Abusnaina et al. propose a Deep Fingerprinting Defender (DFD) against deep WF attacks. DFD injects dummy packets within every burst of Tor traces to generate adversarial traffic traces [41].

IV. Preliminary

Our proposed WF-Transformer utilizes a Transformer network for the traffic temporal feature extraction, rather than commonly used CNN network, which fails to catch the temporal features and the long-range dependence of the traffic trace. In this section, we give some introduction about the Transformer network.

Transformer network is designed with a self-attention mechanism, which is first applied to the field of natural language processing (NLP) [24], [42], [43]. Recently, Transformer network has been applied for computer vision tasks, which is dominant by CNN network previously [25]. Generally, Transformer network contains two parts, an encoder and a decoder. Transformer with encoder-decoder architecture is suitable for machine translation tasks in NLP. For sequence classification tasks, we only need the encoder part of Transformer, and the decoder part can be replaced with a classifier.

The input sequence can be set $X = (x_1, x_2, \ldots, x_n)$. The embedding of the Transformer network is used to extract the features from the original traffic traces, and obtain the vector matrix representation Z of the features. Z can be calculated as follows:

$$Z = W_E X \tag{1}$$

where W_E is the embedding matrix, and different W_E represents different embedding manners.

Behind each block of the Transformer network, there is a layer normalization operation, which is used for the output of the corresponding block, and the outputs are converted to a mean value of 1, and a variance of 0 by the layer normalization.

In the NLP field, the position of different element in a sequence is important for the meaning of a sentence. However, due to the parallelization processing of the Transformer, position embedding is conducted to preserve the position information of a sequence. For a sequence, the position embedding can be calculated as follows:

$$PE(pos, 2i) = sin(\frac{pos}{10000^{\frac{2i}{d_{model}}}}) \tag{2}$$

$$PE(pos, 2i+1) = cos(\frac{pos}{10000^{\frac{2i}{d_{model}}}}) \tag{3}$$

where pos is the position of the corresponding element in a sequence, d_{model} is the word embedding length, and $i \in [1, 2, \ldots, d_{model} - 1]$. When i is an odd number, position embedding is calculated with equation 2, otherwise using equation 3. With the word embedding and position embedding, the composition is obtained by adding the word embedding and position embedding element by element.

The multi-head attention mechanism is an important part of Transformer network, which focuses on key information of the sequence, like a human intuitively. Transformer network utilizes a multi-head attention mechanism to weigh the important features of the sequence. For the embedded features Z, we can get Q, K, and V by linear transform as follows:

$$Q = W_q Z \tag{4}$$

$$K = W_k Z \tag{5}$$

$$V = W_v Z \tag{6}$$

where Q, K, V are query, key and value respectively. W_q, W_k, W_v are initialized randomly and updated in the training procedure.

According the attention mechanism, we can get the attention weight as follows:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{7}$$

where d_k is used for scaling.

V. The Proposed Method

In this section, we first introduce the inputs of our proposed WF-Transformer. Subsequently, we present the architecture of the model and the optimization procedure of our method.

A. The Inputs of WF-Transformer

Generally, Transformer networks are used for NLP, and the input of the Transformer is a sequence, which is processed from the original sentences. As for the traffic traces, the traffic traces collected by network attackers mainly contain the the time stamp of the packets, packet number and packet size and packet directions. Prior study in WF attacks shows that features extracted from the lengths of traces in each direction are important for the success of WF attack. However, Sirinam et al. prove that using packet length as a feature does not provide significant attack accuracy improvement [7]. Their proposed DF attack method abandons the packet size and the time stamps, and only utilizes the packet direction as the input of the deep neural network.

Similar to other deep WF attack methods, we only utilize the packet directions as the inputs of the WF-Transformer. The directions of the packets can be extracted from the collected traces easily, and we use $+1$ to represent the outgoing packet, while -1 is used to represent the incoming packet, and then the input sequence can be donated as $X = [x_1, x_2, \ldots, x_n]$, where $x_i \in \{-1, +1\}$, and $i \in \{1, 2, \ldots, n\}$, n is the number of packets in a trace sequence. The length of the sequence is important for the speed and classification accuracy of the Tor traces, and the sequence length of other deep WF methods [7], [30] are generally set to 5000. In fact, with the help of the Transformer network in temporal feature extraction, WF-Transformer can achieve comparable performance with shorter trace compared with the state-of-the-art deep WF attacks, and the superiority of our proposed method will be shown in the following part.

B. The Architecture of WF-Transformer

Since the CNN focuses on the local property of the sequence and ignores the long-term dependence among the elements in a sequence, CNN has drawbacks in processing time series data, while the Tor trace is a typical time series sequence. In this paper, our proposed WF-Transformer utilizes Transformer network to extract the temporal features of the Tor traces. To make the proposed model suitable for the traffic traces, we conduct sophisticated design on the Transformer network, and we will introduce the design of our model in the following part.

In Figure 2, the architecture of the proposed WF-Transformer network is shown in detail. After the embedding of the inputs, a layer norm operation is conducted,

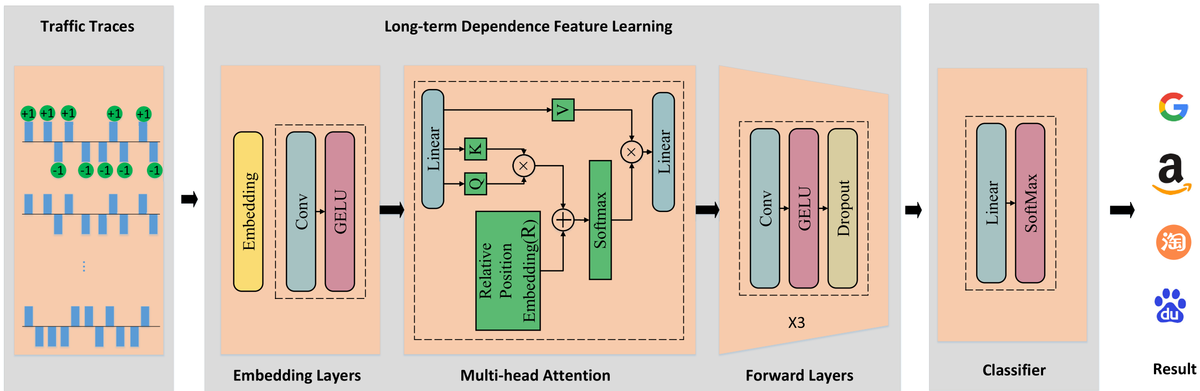


Fig. 2. The architecture of the proposed WF-Transformer.

and then is the multi-head attention module, a one-dimension convolutional network layer with Gaussian Error Linear Unit (GELU) activation function is used for the processing of the embedded features, and the Q, K, V are calculated with corresponding equations (4)(5)(6).

Absolute position embedding is introduced as above, which utilizes a sine or cosine function to generate the position code, and this position embedding fails to distinguish the relationship among multiple elements. In the multi-head attention part of our proposed WF-Transformer, we utilize the relative position embedding manner to show the relative position relationship among different packets in a trace. The multi-head attention with relative position embedding can be calculated as follows:

$$Attention(Q, K, V) = softmax(\frac{Q(K^T + A^K)}{\sqrt{d_k}})(V + A^V) \tag{8}$$

For $a^*_{i,j} \in A^*$, $*$ represents K or V, $a^*_{i,j} = W^*_{clip(j-i,k)}$, and $clip(x, k) = max(-k, min(k, x))$, $k = j - i$. $a^K_{i,j}$ and $a^V_{i,j}$ can represent the relative position information of element i and element j, and this information only related with the position i and j.

Furthermore, we utilize short-cut connection [23] among different layers and dropout function after the GELU activation function to avoid overfitting. In the forward layers, we utilize four convolutional layers, and each with GELU activation. As for the classifier, we utilize a fully-connected layer to classify the Tor trace for our WF-Transformer model.

In this section, we describe the optimization of the proposed WF-Transformer. We use WFT to represent the Transformer feature extractor, and Cls to represent the classifier. Given the input $X = \{x_1, x_2, \ldots, x_m\}$ and ground-truth labels for corresponding samples $Y = \{y_1, y_2, \ldots, y_m\}$ for training set, where m is the number of the samples. C denotes the number of the trace categories. Then we can get the predicted probability $\hat{y}_i$ for the input x_i as follows:

$$\hat{y}_i = softmax(Cls(WFT(x_i))) \tag{9}$$

We choose multi-classification cross-entropy loss function with a smooth coefficient s for our model, and the loss function can be calculated as follows:

$$loss = -\frac{1-s}{|X|}\sum_{i=1}^{|X|}\sum_{c=1}^{C} y_{ic}\log(\hat{y}_{ic}) - \frac{s}{|X|}\sum_{i=1}^{|X|}\sum_{c=1}^{C}\log\left(\hat{y}_{ic}\right) \tag{10}$$

where y_{ic} and $\hat{y}_{ic}$ denote the ground-truth label c and probability of the sample x_i classified to category c, respectively. The smooth coefficient is determined by varying s in the range of $\{0, 0.001, 0.005, 0.01, 0.5\}$ in the experiments, and we set $s = 0.005$ for the proposed model by a grid search manner.

VI. Experimental Evaluation

In this section, we conduct extensive experiments to demonstrate the effectiveness and superiority of our proposed method. As far as we know, our proposed WF-Transformer is the first work that utilizes Transformer network for temporal feature extraction of anonymous traffic traces, and we compare WF-Transformer with the state-of-the-art WF attacks on public Tor traffic datasets on both closed-world and open-world setting.

A. Dataset

We utilize the open-source dataset which is collected by Sirinam et al. [7] to evaluate our method. The dataset contains two parts, Closed-world dataset and Open-world dataset, which is collected with a tor-browser-crawler tool on Tor Browser Bundle version 7.0.6. DF dataset is the latest public dataset for the evaluation of deep WF attacks, which contains 95000 traces while Wang‘s dataset only contains 9000 traces in the closed-world setting, so DF dataset is more suitable for the training and evaluation of deep WF attack models.

1) Closed-World Dataset: This part of dataset collects Tor traffic traces by visiting the homepage of each top Alexa 100 websites 1250 times. This dataset is generated in a campus environment by using ten low-end machines. These machines visit target websites with sequential and ordered methodology in a round-robin manner. The dataset is preprocessed after the crawler finishes by only recording the websites that have over 1000 visits. The closed-world dataset contains 95 websites with 1000 visits for the closed-world evaluations. For closed-world evaluation, the training set contains 76000 traces, while the testing set contains 9500 traces.

TABLE I
HYPERPARAMETER SELECTION FOR WF-TRANSFORMER

Hyperparameters	Search Range	Final
Input Dimension	[500,...,7000]	1000,2000,3000,4000,5000
Embedding Size	[64, 128, 256, 512]	128
Optimizer	[Adam, Adamax, RMSProp, SGD, AdamW]	AdamW
Learning Rate	[0.001 ... 0.01]	0.0003
Training Epochs	[10 ... 50]	40
Mini-batch Size	[16 ... 256]	16
Output Dimension of K, V	[128, 256, 512, 1024]	512, 512
Parameters of Convs		
Output channel of Conv#1	[2 ... 16]	3
Kernel size of Conv#1,2,3,4	[2 ... 16]	9
Stride of Conv#1,2,3,4	[0,1,2,3,4]	1
Padding of Conv#1	[0,1,2,3,4]	4
Output channel of Conv#2,3,4	[16, 32, 64, 128, 256, 512]	512
Padding of Conv#2,3,4	[0,1,2,3,4]	0
Activation Functions	[Tanh, ReLU, ELU,GELU]	GELU, ReLU
Pooling Layers	[Average, Max]	Max
Number of FC Layers	[1 ... 4]	2
Hidden units (each FCs)	[256 ... 2048]	[512, 512]
Dropout [Attention, FC1, FC2]	[0.1 .. 0.8]	[0.1, 0.1, 0.1]

TABLE II
THE DETAILS OF DF DATASET

Dataset	# of websites	# of traces/website	Length of trace	# of traces (training set)	# of traces (testing set)
Closed-world dataset	95	1000	5000	76000	9500
Open-world dataset	40716	1	5000	96000	29500

2) Open-World Dataset: This part of the dataset collects the Alexa top 5000 websites excluding the first 100 websites used in closed-world dataset. The collection procedure is the same as the closed-world collection procedure. For open-world dataset collection, each homepage of the websites is only visited once, and the corrupted visits which have been collected in closed-world dataset are discarded. The webpages that are blank pages, access denied pages, CAPTCHA pages and timeout error pages are removed, and the final open-world dataset contains 40,716 traffic traces. For open-world evaluation, the training set contains 96000 traces, while the testing set contains 29500 traces. More details can refer to Table II.

B. Experiment Details

Our proposed WF-Transformer method is used for website fingerprinting attacks with a novel deep network architecture. Compared with existing deep website fingerprinting attack methods which utilize CNN as the feature extractor, the proposed WF-Transformer utilizes Transformer which contains lots of improvements for traffic trace feature extraction.

Our implementation of WF-Transformer model uses Pytorch framework [44], and we utilize RTX 3090 to accelerate the model training. We utilize AdamW [45] as an optimizer to train the WF-Transformer networks, the learning rate is set as $3e^{-4}$ at the beginning with a dynamic change mechanism. The learning rate is $3e^{-4}$ in the first 30 epochs, and then set as $3e^{-5}$ for the following 10 epochs. The batch size is set to 16, and we utilize the dropout layer after each activation layer and set the rate as 0.5, and the training epochs are set to 40 for all the attacks. The input of the WF-Transformer is the direction of each packet in the traffic trace. Furthermore, we give hyper-parameter search ranges and the selected values in Table I.

C. Experiment Setup

To evaluate the performance of our proposed WF-Transformer, we compare it with several state-of-the-art WF attacks. We choose three attacks, CUMUL [13], AWF [29] and DF [7] as benchmarks to show the performance of our proposed attacks. CUMUL is a non-deep WF attack, AWF and DF are deep WF attacks with different designs of CNN networks, and they are two kinds of representative existing WF methods. Specifically, we follow the network setting of AWF [29], AWF is an LSTM/CNN hybrid architecture in our experiment evaluations, which is built by five convolutional layers, one LSTM layer and a fully-connected layer.

To evaluate the performance of our proposed WF-Transformer comprehensively, we choose four defenses, TAMARAW [16], WTF-PAD [17], FRONT [35] and Mockingbird [40] as competitors to our proposed attacks. TAMARAW is a heavyweight regularization defense with high latency and data overhead, while WTF-PAD is a lightweight obfuscation defense, which is developed by Tor project. FRONT is a new lightweight defense which pads dummy packets into the front part of the traces. Mockingbird generates adversarial traces to confuse the classifiers. In our experiments, we follow the original settings that provided in the papers and perform parameter tuning on the candidate parameters and find the optimal parameters. Specifically, Mockingbird utilizes the trace burst as the input, so we transform the burst of the generated adversarial trace to the direction of each packet for the evaluation of WF attacks.

We conduct the experiments on both closed-world and open-world settings. In closed-world setting, the classification accuracy is used for the evaluation of the attack methods. In open-world setting, we calculate the corresponding TPR, FPR and precision-recall curves to show the performance of different methods.

D. Evaluation Metrics

We use the prediction probability output by the classifier to classify target traces. If a monitored trace has a prediction probability greater than a certain threshold, we will record it as True Positive (TP), or False Negative (FN) otherwise. Similarly, if an unmonitored trace is correctly classified, it will be considered True Negative (TN), or False Positive (FP) otherwise.

For closed-world scenario, classification accuracy is used for the evaluation of WF attacks. For open-world scenario, True Positive Rate (TPR) and False Positive Rate (FPR) are used. Furthermore, Precision-Recall curves are used for further study of the WF attacks for open-world setting. As for closed-world setting, the classification accuracy is calculated by N_{ACC}/N_{ALL}, where N_{ALL} is the number of the all traffic traces, and N_{ACC} denotes the number that correct classifications. TPR is calculated by $TP/(TP + FN)$ and FPR equals to $FP/FP + TN$. Precision and recall are defined as $TP/(TP + FP)$ and $TP/TP + FN$, respectively.

VII. Experiment Results

In this section, we show the experiment results of our WF-Transformer for both closed-world setting and open-world setting.

A. Closed-World Evaluation on Non-Defended Traces

We evaluate the performance of the WF-Transformer in the closed-world scenario on the non-defended traces, which only utilizes website traces from the closed-world dataset with no WF defenses. For baseline methods, we compare WF-Transformer with CUMUL, AWF and DF attacks. As mentioned ahead, WF-Transformer is designed for the temporal feature extraction of the traffic traces. In the evaluation phase, unlike the setting in the prior attacks, we vary the length of the input trace and show the superiority of the proposed WF-Transformer.

Table III shows the classification accuracy results of different WF attacks, and we repeat each anonymous trace classification task five times. Both average classification accuracy and standard deviation are reported for each anonymous trace classification task. In table III, we show the results of different lengths of inputs for each WF attack. Generally, the length of inputs in prior deep WF attacks is set to 5000, such as AWF and DF attacks. However, the longer the inputs, the higher the model computation cost is, and the practicality of the model can be reduced, since attackers need to collect more traffic traces, which needs more time. The length of the inputs is set in the range of [1000, 2000, 3000, 4000, 5000] for each attack.

According to Table III, WF-Transformer attains 99.1% classification with the input length 5000, which is significantly better than the other attacks with different input lengths, significantly. Furthermore, we can find that the classification accuracy is improved with the length of the inputs increased. Compared with CUMUL, which is a non-deep WF attack method, our WF-Transformer outperforms CUMUL in all input length settings, significantly.

Compared with deep WF attacks, such as AWF and DF methods, WF-Transformer can still obtain better performance than them, especially for the short input length setting. In our experiment, AWF utilizes the CNN/LSTM hybrid architecture for the model building, which is the same as the corresponding hyperparameter setting for the AWF. We can observe that the classification accuracy of AWF is lower than that of our WF-Transformer, even when the input length is set to 5000, and the classification accuracy of AWF (95.5%) is poorer than the performance of WF-Transformer with input length 1000. Though AWF model contains LSTM network to extract temporal features of the traces, the performance of AWF is still inferior to our proposed WF-Transformer, which demonstrates the superiority of the WF-Transformer in trace temporal feature extraction. DF has a more complex network architecture than AWF, and gets better performance than AWF, but the performance of DF is still inferior to WF-Transformer. When the input length of the WF-Transformer is set as 2000, WF-Transformer gets better classification than DF with input length 5000 (98.7% vs 98.3%), which demonstrates the effectiveness and superiority of our proposed WF-Transformer for anonymous trace classification in closed-world scenario. Furthermore, based on the given average classification accuracy and standard deviation, a t-test [46] can be conducted to each anonymous trace classification result, there is high confidence that our WF-Transformer significantly outperforms DF for corresponding input trace length. For the case that WF-Transformer with input length 5000 and DF with input length, the classification accuracy of our proposed WF-Transformer is significantly higher than that of DF.

B. Closed-World Evaluation on Defended Traces

In this section, we test the performance of the proposed WF-Transformer on the defended traces. As mentioned in section III, WF defenses pad dummy packets into the normal traces to change the pattern of the anonymous traces, so that the performance of WF attack model in classifying defended traces is reduced. In this part, we select TAMARAW [16], WTF-PAD [17], FRONT [35] and Mockingbird [40] to evaluate the performance of WF attack under the defended traces. TAMARAW is a heavyweight defense that pads dummy packets into traces at fixed speed and length, which causes

TABLE III
CLASSIFICATION ACCURACY (%) ON NON-DEFENDED TRACES FOR STATE-OF-THE-ART ATTACKS IN THE CLOSED-WORLD SETTING

Attacks \ Length of inputs	1000	2000	3000	4000	**5000**
CUMUL [13]	$93.8^{\pm 0.2}$	$95.1^{\pm 0.1}$	$95.2^{\pm 0.1}$	$95.5^{\pm 0.1}$	$95.5^{\pm 0.1}$
AWF [29]	$92.3^{\pm 0.2}$	$94.4^{\pm 0.1}$	$95.6^{\pm 0.1}$	$95.7^{\pm 0.1}$	$95.7^{\pm 0.1}$
DF [7]	$96.4^{\pm 0.1}$	$97.3^{\pm 0.1}$	$98.1^{\pm 0.1}$	$98.3^{\pm 0.1}$	$98.3^{\pm 0.1}$
WF-Transformer (Ours)	$97.8^{\pm 0.1}$	$98.7^{\pm 0.1}$	$98.9^{\pm 0.1}$	$99.0^{\pm 0.1}$	$\mathbf{99.1}^{\pm 0.1}$

high bandwidth overhead and delay. WTF-PAD and FRONT are lightweight defenses. According to the experiment in the closed-world setting, WF-Transformer can utilize short inputs to get better classification accuracy than other methods, so evaluating performance on the traces defended by FRONT is necessary for our proposed WF-Transformer. Mockingbird is a new defense that based on adversarial examples, we test the performance of our proposed WF-Transformer on the traces defended by Mockingbird.

We follow the setting of TAMARAW in the original paper, and padding dummy packets into the traces with fixed IAT. For WTF-PAD defense, we use an open-source defended dataset. For FRONT defense, since there is no time stamp in our dataset, and we follow the setting of the original paper [35] and calculate the percentage of packets in the traces, and we only pad dummy packets into first 50% of the traces to get the traces defended by FRONT.

Since the classification accuracy of WF-Transformer with input length 2000 on non-defended traces has reached 98.5%, which is higher than other WF attacks, the following evaluation of WF-Transformer on defended traces is based on the input length 2000, and for other WF attacks, the input length is set to 5000 as the original paper. In our experiment settings, the defended traces are also used for the training of the attack models, and we evaluate the classification accuracy on test traces. For Mockingbird [40], we utilize AWF network as the detector, and the experiment set follows the case I in the original paper [40], which is more realistic.

We show the defense performance in Table IV, and both the classification accuracy on defended traces and bandwidth overhead are listed in the table. TAMARAW is a heavyweight defense, and the bandwidth overhead is 403% in our experiments, and holds up well with less than 15% classification accuracy among all the WF attacks. Though TAMARAW gets good performance in defending the WF attacks, it is not practical in real anonymous networks due to the high bandwidth overhead and delay.

For WTF-PAD defense, the classification accuracy of CUMUL and AWF is reduced from over 90% to less than 65%, which shows significant defense performance. However, DF can still realize over 90% classification accuracy on WTF-PAD defended traces (98.3% for non-defended traces), and the experiment result demonstrates that WTF-PAD fails to resist the DF attack. Furthermore, the proposed WF-Transformer obtains 92.1% classification accuracy on defended traces, which shows that WTF-APD defended traces fail to confuse the WF-Transformer classifier. In our experiment, the bandwidth overhead of WTF-PAD is 67%, and there is no delay for WTF-PAD defense due to the dummy packets being padded into the gaps between normal packets.

In Table IV, we can observe that FRONT defense can reduce the classification accuracy of all the WF attacks, and the accuracy of CUMUL is reduced from 95.5% to 44.2%. For AWF and DF defenses, FRONT still gets good performance in confusing deep classification models, and the classification accuracy is lower than 72% on the defended traces. According to the FRONT paper [35], the trace pattern information is gathered in the head part of the traces, padding dummy packets into the front part of the traces rather than the whole trace can reduce the bandwidth overhead and confuse traffic pattern information efficiently. Since the feature-rich part is confused, the attack performance is reduced. For Mockingbird defense, WF-Transformer gets 56.2% classification accuracy on the traces defended by Mockingbird, which is higher than other WF attacks. The experiments on defended traces show that our WF-Transformer can get better performance on defended traces than state-of-the-art methods.

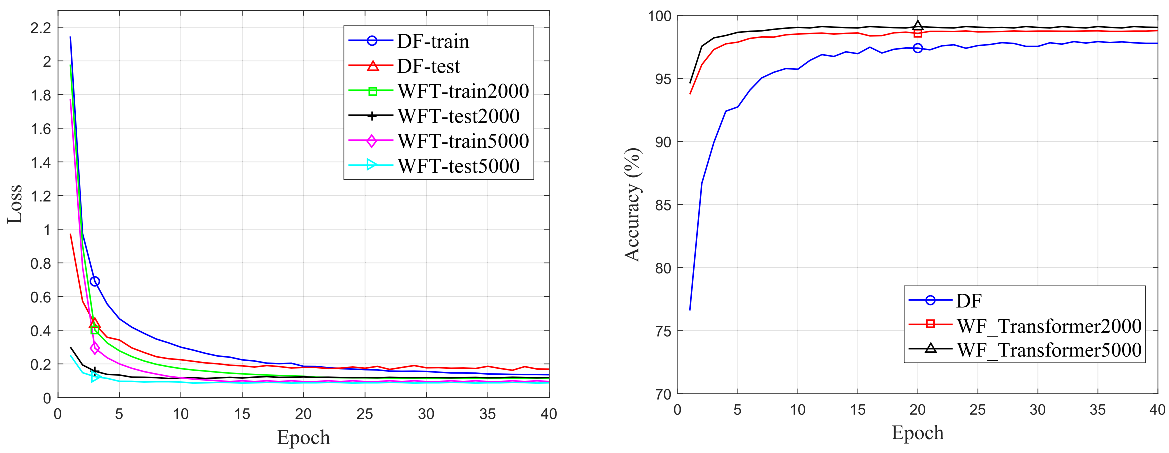
Our proposed WF-Transformer gets better attack performance than AWF and DF attacks, though only using 2000 cells in a trace. WF-Transformer can extract the temporal feature of the traces, and the classification accuracy of WF-Transformer is reduced from 98.5% on non-defended traces to 82.4% on traces defended by FRONT, which gets the smallest drop compared with other WF attacks.

Furthermore, we show the training process of the WF-Transformer in undefended traces in a closed-world setting. Since DF is a state-of-the-art deep WF attack, we compare our method with DF to show the convergence performance. As shown in Figure 3, we show the DF method with 5000 input length and our proposed WF-Transformer with two kinds of input length, 2000 and 5000, respectively. Figure 3(a) shows changes in the training and test loss during training epochs. WF-Transformer method has lower training and test loss value than DF method, and we can find that the test loss value of our proposed WF-Transformer is almost unchanged after 5 epochs, while the loss of DF remains unchanged after 10 epochs, which means that our proposed WF-Transformer converges faster than DF methods. Furthermore, WF-Transformer with input length 5000 has a lower loss than WF-Transformer with length 2000 for the training and test procedure.

In Figure 3(b), we show the classification accuracy of DF and WF-Transformer during training epochs. Our proposed WF-Transformer obtains over 90% classification accuracy,

TABLE IV
CLASSIFICATION ACCURACY (%) ON DEFENDED TRACES IN THE CLOSED-WORLD SETTING

Defense	Accuracy (%) of WF attacks on defended traces				Bandwidth overhead (%)	Delay
	CUMUL [13]	AWF [29]	DF [7]	WF-Transformer (Ours)		
TAMARAW [16]	14.1	11.5	12.6	14.7	401	Yes
WTF-PAD [17]	62.4	61.2	90.4	92.1	67	No
FRONT [35]	44.2	51.2	71.1	82.4	79	No
Mockingbird [40]	19.0	36.2	42.0	56.2	58	Yes



(a) Closed-world: Loss value curve

(b) Closed-world: Classification accuracy curve

Fig. 3. The loss value and classification accuracy change for DF and WF-Transformer in closed-world setting. (WFT denotes WF-Transformer, the number represents the input length.)

while DF gets about 76% classification accuracy in the first epoch, and the classification accuracy of WF-Transformer is higher than that of DF for each training epoch, consistently. The classification accuracy of WF-Transformer has little change after 5 epochs, while the accuracy for DF remains stable after 15 epochs, which shows that WF-Transformer has better convergence than DF. Furthermore, since we update the learning rate dynamically in the training, the loss and accuracy curves of WF-Transformer are smoother than that of DF.

C. Open-World Evaluation

WF attacks used in the open-world scenario are more realistic, in which attackers not only classify the anonymous traces based on the limited monitored sites, but also need to distinguish whether the trace comes from a monitored site or an unmonitored one. For the evaluation of the WF attacks in the open-world setting, we chose the true positive rate (TPR) and false positive rate (FPR), and we also show Precision-Recall (PR) curves, which is commonly used in prior works [7], [44].

In our open-world evaluation, the output of the model is the prediction probability of each category. Previous studies have provided a *Standard Model* for the evaluation of WF in the open-world setting [7], in which the attackers can utilize the unmonitored traces to train the classifier, and this can help the WF attack model classify monitored and unmonitored traces correctly. The evaluation of the standard model is similar to that in the closed-world setting, the difference is that we add an extra category to the classifier for the classification of the unmonitored website traces. Generally, if the prediction probability of a monitored trace is greater than a threshold, the trace is considered a true positive. The different thresholds are used for different WF attacks, which are selected to realize high TPR and low FPR.

For the evaluation in the standard model, we investigate the impact of more training data on the performance of the classifiers in the open-world scenario by varying the number of unmonitored traces, and we compare the performance of our proposed WF-Transformer with other state-of-the-art WF attacks on non-defended traces. In our standard model, the training set contains all the monitored traces(900 traces for each 95 monitored websites). In the test phase, all the monitored traces and 20000 unmonitored traces are included in the test set, and the unmonitored traces are randomly selected from the unmonitored traces set that is not used for training.

As shown in Figure 4, we show the influence of the unmonitored traces used for the training on the WF attacks in open-world setting. For the evaluation of WF-Transformer, we show the results on two settings, the input length is 2000 and 5000, respectively, and the amounts of the unmonitored traces vary in the range {1000, 5000, 10000, 15000, 20000}, Figure 4 (a) and (c) show the TPR and FPR of the baseline methods and the proposed WF-Transformer with input length 2000, and Figure 4 (b) and (d) show the TPR and FPR of the baseline methods and the proposed WF-Transformer with input length 5000.

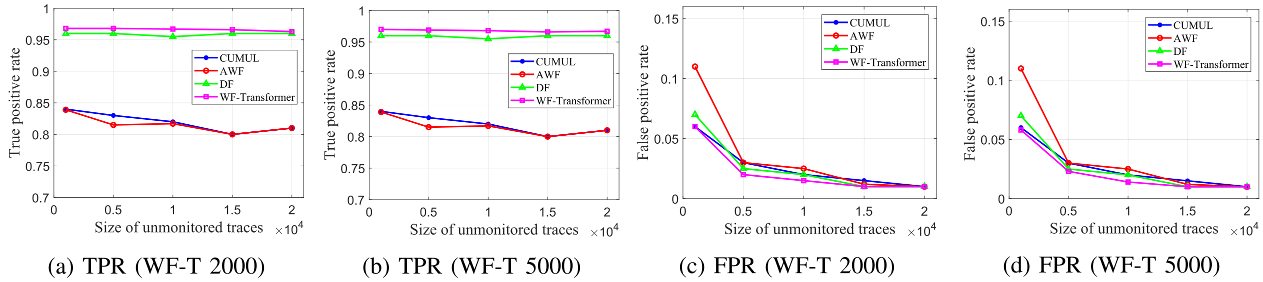


Fig. 4. The impact of the amount of unmonitored training data on TPR and FPR (Non-defended dataset) in open-world setting. (WF-T represents“WF-Transformer”. 2000 and 5000 represent the input length for WF-Transformer, respectively.)

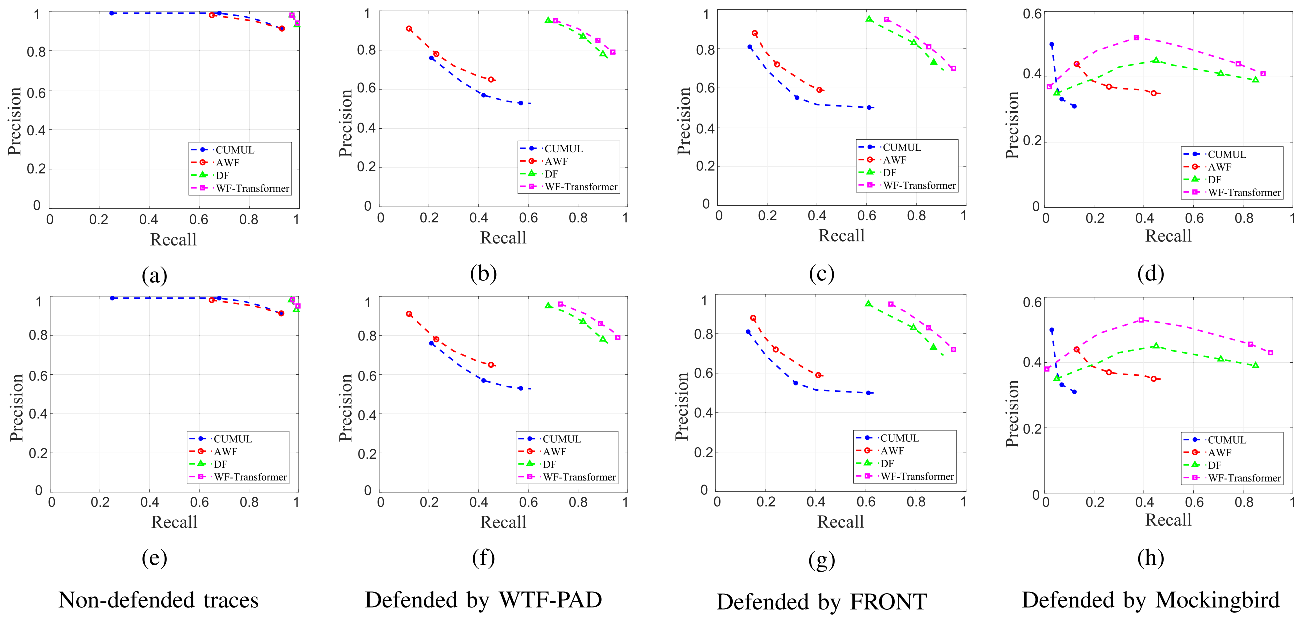


Fig. 5. Precision-Recall curves in the open-world setting. (Up: the input length of the traces for WF-Transformer is set to 2000. Down: the input length of the traces for WF-Transformer is set to 5000.)

Compared with the baseline methods, WF-Transformer gets the best performance in both TPR and FPR metrics. For the input length in 2000, WF-Transformer gets consistently best performance on both TPR and FPR, and the TPR is 0.965, while FPR is 0.007 for 20000 unmonitored traces. For the input length in 5000, WF-Transformer still gets best performance than the other baseline methods, and the TPR is 0.969 and the FPR is 0.007 for 20000 unmonitored traces. According to our experiments, AWF gets the lowest TPR and highest FPR, which is a bad performance on unmonitored traces, and this be attributed to the architecture of the networks, which has fewer layers than DF. WF-Transformer and other baseline methods have the same FPR trend with the training size increasing, while our proposed WF-Transformer has higher TPR than other baseline methods over all the training sizes, which demonstrates the effectiveness and superiority of the proposed WF-Transformer.

For the evaluation of the unmonitored traces in the open-world scenario, we conduct experiments on the traces defended by WTF-PAD and FRONT methods. Furthermore, we also conduct experiments on the non-defended traces for comparison. According to Figure 4 (c) and (d), the FPR drops with the size of the unmonitored trace increasing in the training, and the FPR is lowest when the size is set to 20000. In the evaluation of the defended traces, we set the number of the unmonitored trace to 20000 and provide Precision-Recall curves of different WF attacks to show the diagnostic ability when the discrimination threshold varies. In the experiment setting, the training set contains all the monitored traces and 20000 unmonitored traces, while the test set contains the monitored set and 20000 unmonitored traces (each unmonitored website has one trace).

We conduct extensive experiments in an open-world setting to show the precision-recall curves in the non-defended, defended by WTF-PAD and defended by FRONT traces. The precision-recall curve shows the trade-off between precision and recall for different thresholds. A high area under the curve represents both high recall and high precision, where high precision relates to a low false positive rate, and high recall relates to a low false negative rate. Since the dataset we used is an imbalanced dataset [7], PR curves can be used to show the performance of the classifier. As shown in Figure 5, we show the PR curves in the open-world settings under different input length settings for WF-Transformer, in which the input length for CUMUL, AWF, and DF is set to 5000 as their original paper, while the input length for WF-Transformer is set to 2000 and 5000, respectively.

The up row in Figure 5 shows the PR curves of the CUMUL, AWF, DF and WF-Transformer with input length 2000 on the defended traces, and we can observe that WF-Transformer still outperforms other WF attacks in all the defense settings. For non-defended traces, the PR curves of DF and WF-Transformer are in the top right-hand corner, which means that both DF and WF-Transformer are effective for different thresholds, and the performance of DF and WF-Transformer are close to each other in the non-defended traces. The precision for CUMUL and AWF is high, while the recall covers a wide range, which means that these attacks miss many monitored traces. For the traces defended by WTF-PAD, the performance of all the attacks is reduced. According to the Figure 5(b), CUMUL and AWF get bad performance for the traces defended by WTF-PAD, while our proposed WF-Transformer outperforms all other WF attacks, including DF attacks. For traces defended by FRONT, our proposed WF-Transformer still gets the best performance than other attacks, and for high precision, WF-Transformer gets a precision of 0.968 and recall of 0.711. For high recall, WF-Transformer gets 0.720 precision and 0.956 recall. With the input length set to 2000, WF-Transformer gets best performance than other WF attacks, which set the input length of the trace to 5000.

In Figure 5 (d) and (h), we show the performance of WF-Transformer on traces defended by Mockingbird method. Mockingbird is a new defense that generates adversarial traces to mislead the classifiers to a selected wrong target category [40], and makes the performance of WF attack decrease. For traces defended by Mockingbird, our proposed WF-Transformer gets the best performance than other attacks, and for high precision, WF-Transformer gets a precision of 0.562 and a recall of 0.391. For high recall, WF-Transformer gets 0.425 precision and 0.911 recall. The area under the PR curve of WF-Transformer is larger than other WF attacks, which means that WF-Transformer has better anonymous trace classification performance. In Figure 5 (d), WF-Transformer with the input length set to 2000 still gets the best performance than other WF attacks which the input length of the trace is set to 5000.

The down row in Figure 5 shows the PR curves of the CUMUL, AWF, DF and WF-Transformer with input length 5000 on the defended traces. With more information delivered to WF-Transformer model, our proposed method gets better performance than other WF attacks and WF-Transformer with input length 2000. WF-Transformer can utilize short trace (2000) to train the model and get better performance than other deep attacks trained on long traces (5000) in both closed-world and open-world settings.

VIII. Measuring Information Leakage for Extracted Features

In this section, we give more evidence to show the superiority of our proposed WF-Transformer in extracting trace features. As we claimed that our proposed WF-Transformer can learn the temporal features of the traces, which is ignored by the CNN-based deep WF attacks. There are several studies about the information leakage measurement for website fingerprinting [47], [48], and the information leakage measurement can be used to estimate how much information leaks from systems.

We utilize WeFDE (**We**bsite **F**ingerprint **D**ensity **E**stimation) [48] to measure the information leakage of the features extracted by DF and our proposed WF-Transformer. According to WeFDE, the information leakage can be calculated as $I(F; W) = H(W) - H(W|F)$, where F is the trace fingerprinting, and W is the website information, and $I(F; W)$ is the amount of the information that an attacker can learn form F about W, and $H(\star)$ is entropy function. Since WeFDE is specially designed for quantifying the information leakage from WF defenses [48], we measure information leakage of the features extracted from the traces defended by WTF-PAD.

We utilize the features which are before the classifier of WF-Transformer and DF as inputs of WeFDE, and implement the source code[1] to calculate the information leakage.

According to our experiment, the information leakage of features extracted by DF (input length 5000) is 7.2 bits, while the information leakage of features extracted by WF-Transformer is 7.7 bits (input length 2000) and 7.9 bits (input length 5000). The higher of the information leakage, the more information is extracted from the traces. The experiment result demonstrates that our proposed WF-Transformer can learn more information from the anonymous traces than the CNN-based model, which validates the superiority of our proposed method.

IX. Discussion

To the best of our knowledge, WF-Transformer is the first work that uses Transformer network for WF attack tasks. According to the experiment results above, our proposed WF-Transformer obtains the best performance than several state-of-the-art WF attacks on both closed-world and open-world settings. Furthermore, we conduct extensive experiments to show the model robust on the defended anonymous traces, and the experiment results demonstrate the effectiveness and superiority of our proposed method.

As we have claimed that our proposed WF-Transformer can extract the temporal feature of the traces with the help of the Transformer network, WF-Transformer can learn the long-term dependence among the trace cells, while the CNN network focuses on the local features with a slide window and ignores the dependence between the front cells and the latter cells in the traces, and Transformer architecture has advantages in analyzing trace data. Though the proposed WF-Transformer obtains convincing results in WF attack, we give some consideration to the limitation of our proposed method, and the discussion mainly focuses on the training costs.

We test the processing time for the WF attacks with different deep networks. For simplicity, we compare the processing time

[1] https://github.com/s0irrlor7m/InfoLeakWebsiteFingerprint

TABLE V
THE COMPUTING TIME FOR DIFFERENT DEEP WF ATTACKS. (SECOND/TRACE)

Method \ Length	2000	5000
DF [7]	0.0026	0.0027
WF-Transformer (OURS)	0.007	0.019

of our proposed WF-Transformer with DF attack, which is a state-of-the-art attack based on CNN architecture. Based on the DF original code, we reproduce the DF with Pytorch framework, and WF-Transformer is realized with Pytorch, equally. Both of them use GPU acceleration with NVIDIA GTX 3090 with 24 GB of GPU memory. For the input length of 2000, DF requires 0.0026 seconds for each trace, while WF-Transformer requires 0.0070 seconds for each trace. For the input length of 5000, the processing time for each trace has small increase, and requires 0.0027 seconds, while WF-Transformer requires 0.019 seconds. According to the comparison, the time consumption for WF-Transformer is higher than DF method, which is based on CNN architecture. The comparison of computing time is shown in the table V.

The performance of WF-Transformer with a 2000 input length is better than DF with an input length of 5000, the processing time of WF-Transformer for each trace is about 7 times that of DF. Considering the whole attack procedure, collecting only 2000 cells is easier and time-saving than collecting 5000 cells for a website, so the increased time consumption in classification can be tolerated. According to convergence analysis in Figure 3, WF-Transformer converges faster than DF, which can decrease the training time and improve deployment efficiency in real networks. Furthermore, Transformer architecture has been extensively studied in multiple fields, such as NLP [24]and CV [25]. Lots of optimization on Transformer architecture has been proposed, such as Swin-Transformer [25], so it is promising to improve the efficiency of WF-Transformer to decrease the time cost in processing traces, and make our proposed method more efficient.

X. CONCLUSION

In this paper, we present a novel deep WF attack, called Website Fingerprinting Transformer (WF-Transformer), which utilizes Transformer network to extract the temporal feature of the anonymous traffic traces, and trains the model in an end-to-end manner. Unlike existing deep WF attacks based on CNN architecture that only focus on the local dependence of the traces, WF-Transformer can learn the long-term dependence among the trace cells. We conduct extensive experiments to validate the effectiveness of our proposed method in both closed-world and open-world settings. According to our experiments, WF-Transformer outperforms other Bstate-of-the-art WF attacks in both closed-world and open-world setting, and the performance of WF-Transformer with input length 2000 outperforms that of DF with input length 5000 due to Transformer network can capture the temporal feature from the traces, and WF-Transformer with input length 5000 obtain 99.1% classification accuracy for undefended trace, which is a new state-of-the-art WF result. With the input length set to 2000 for WF-Transformer, in closed-world evaluation, WF-Transformer obtains 98.7% classification accuracy for undefended traces, and reaches 92.1% and 82.4% classification accuracy for traces defended by WTF-PAD and FRONT respectively, which is higher than that of DF significantly. For open-world evaluation, WF-Transformer with 2000 input length obtains a 0.99 precision and a 0.95 recall for undefended traces, and it attains precision of 0.97, 0.95 and recall of 0.71, 0.70 for traces defended by WTF-PAD and FRONT, respectively. For the defense method based on adversarial examples, WF-Transformer still gets the best classification performance than existing WF attacks. Furthermore, WF-Transformer has better convergence performance than DF, and these experiments demonstrate the effectiveness and superiority of our proposed WF-Transformer. In the future, we plan to conduct further research on the design of Transformer network for processing traffic traces, and get a more efficient Transformer-based WF model.

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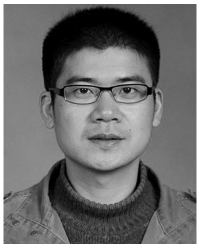

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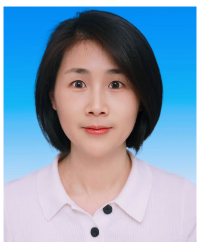

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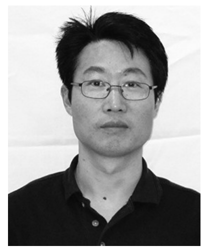

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